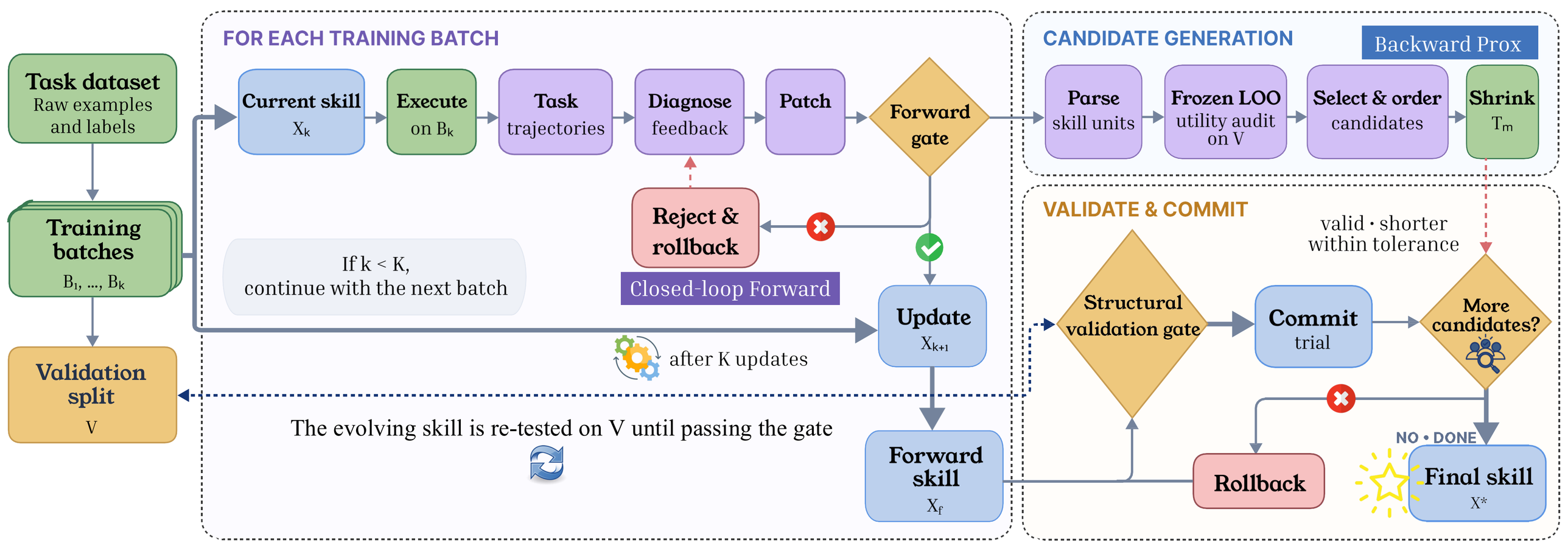

Task dataset
Raw examples and labels
Training batches
$B_1, ..., B_k$
Validation split
V
FOR EACH TRAINING BATCH
Current skill
X_k
Execute
on B_k
Task
trajectories
Diagnose
feedback
Patch
Forward gate
Reject & rollback
Closed-loop Forward
If k < K,
continue with the next batch
Update
X_{k+1}
after K updates
The evolving skill is re-tested on V until passing the gate
Forward skill
X_f
CANDIDATE GENERATION
Backward Prox
Parse
skill units
Frozen LOO
utility audit on V
Select & order
candidates
Shrink
T_m
VALIDATE & COMMIT
valid · shorter
within tolerance
Structural validation gate
Commit
trial
More candidates?
Rollback
NO • DONE
Final skill
X*